

Personal Statement

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UK Global Talent Visa — Digital Technology (Exceptional Promise)

I am an independent researcher and systems engineer based in Derby, United Kingdom. I hold a Bachelor's degree in Computer Science and Engineering, and my work sits at the intersection of Scientific Computing, Computational AI, Cyber Security, and financial technology. Over the past three years—without institutional affiliation, funding, or supervision—I have built production-grade compliance infrastructure, developed original machine-learning methods, filed UK intellectual property, and produced public research with practical applications in digital assets, financial stability, and critical infrastructure.

I am applying under the Exceptional Promise route because my track record shows both technical depth and strong potential for commercial impact in the UK digital technology sector.

1. AMTTP: pre-settlement compliance infrastructure

My principal project is the Anti-Money Laundering Transaction Trust Protocol (AMTTP), a blockchain-native compliance system designed to make risk decisions before transaction finality. This is the central product insight behind the platform. Most existing compliance tools in digital assets operate after settlement, when funds have already moved and risk has already crystallised. AMTTP instead performs machine-learning risk scoring, graph-based relationship analysis, sanctions screening, and policy evaluation within the Ethereum confirmation window (approximately 12 seconds), allowing a transaction to be approved, reviewed, escrowed, or blocked before settlement.

I built AMTTP as a four-layer architecture: open-source TypeScript and Python SDKs; a compliance layer of 7 parallel microservices; a machine-learning pipeline trained on 2.64 million Ethereum transactions; and a blockchain layer of 18 Solidity smart contracts. The production ML system achieved ROC-AUC 0.9879, and the wider platform runs across 17 containerised services with API gateway security, cryptographic signing, rate limiting, and monitoring. I also developed a cross-platform Flutter application, a Next.js “War Room” dashboard, and wallet integrations for real-time user use.

2. Privacy-preserving and cyber-secure design

I developed zkNAF, a zero-knowledge compliance framework, to address the conflict between regulatory transparency and privacy. It enables sanctions verification without revealing identities, risk validation without exposing raw scores, and KYC confirmation without disclosing unnecessary personal data. This is important both commercially and technically: it aligns with GDPR-style data minimisation while improving cyber-security

by reducing data exposure, strengthening auditability, and making compliance flows more tamper-resistant.

3. Research foundation: Blind Spot Decomposition Theory

Alongside the engineering work, I developed Blind Spot Decomposition Theory (BSDT), a machine-learning framework for identifying hidden risk factors that ordinary detectors miss. BSDT addresses a practical problem in fraud and anomaly detection: models often fail not because they are useless, but because important risk information is concealed, weakly represented, or structurally indirect. My framework provides a formal way to identify these blind spots and improve detection without retraining the underlying model. In validation, BSDT improved recall by up to 84.4 percentage points across multiple datasets and domains. This research now underpins the analytical logic of the AMTTP risk engine.

4. Cross-domain system stability

I extended the same computational framework beyond fraud detection into broader system-stability analysis. The same approach can be used for financial-system risk, blockchain transaction monitoring, infrastructure instability, and other complex dynamic systems. This matters because it shows that the work is not a one-off compliance tool, but a reusable computational method for early failure detection. For example, the framework identified structural instability in the ERCOT electricity crisis ahead of cascading failure, showing relevance to critical infrastructure as well as digital finance.

5. Intellectual property, open source, and commercial potential

On 4 March 2026, I filed a UK patent application comprising 25 claims covering machine-learning risk modelling, blockchain compliance systems, and cross-domain stability analysis. I have also open-sourced key parts of the stack, including TypeScript and Python SDKs, machine-learning pipelines, smart contracts, and systemic-risk tools. This combination of patent protection and public technical output reflects both originality and practical execution.

The commercial opportunity is clear. Current on-chain compliance providers such as Chainalysis, Elliptic, and TRM Labs operate mainly post-settlement. AMTTP is designed as pre-trade compliance infrastructure for regulated institutions and Virtual Asset Service Providers entering decentralised finance. The system already supports batch scoring up to 10,000 transactions, is deployed on Ethereum Sepolia for validation, and is designed around FCA-facing and FATF Travel Rule requirements. In commercial terms, this is a credible Compliance-as-a-Service product for enterprise fintech.

6. Why the UK

The UK is the right environment for this work because it combines fintech leadership, strong regulatory institutions, and world-class technical research. My work aligns directly with UK priorities in digital-asset compliance, financial stability, cyber security, and applied AI. My immediate plan is to continue publishing the research, develop AMTTP into a market-ready platform for regulated institutions, and build a UK-based company focused on machine learning, compliance infrastructure, and resilient digital systems.

7. Closing statement

What distinguishes my profile is the ability to combine computer-science training, original research, and production-grade system design. Working independently, I have built a full compliance protocol with 17 microservices and 18 smart contracts, trained ML systems on 2.64 million blockchain transactions, developed original methods with cross-domain relevance, filed a UK patent application, and open-sourced core components of the platform. I believe I can contribute meaningfully to the UK's digital economy as a computer scientist building secure, intelligent, and commercially relevant systems, and I am seeking the opportunity to continue that work in the United Kingdom.